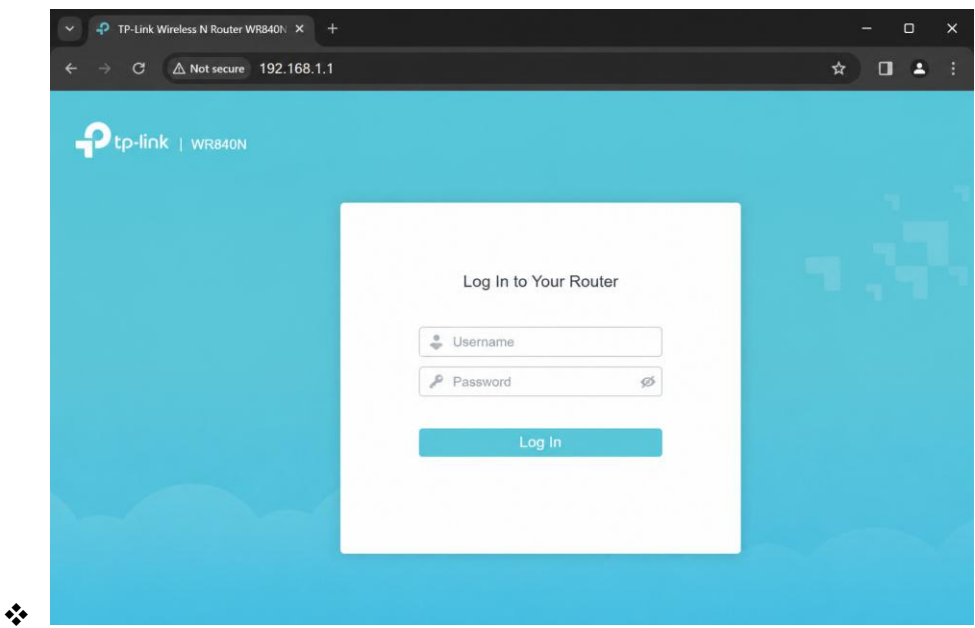


Configuring Basic Network Hardware Theory

- 1) Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



- 2) Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Parameter	Details
IP Address	192.168.1.105
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

- 3) Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

❖ **Before Change (DHCP):**

- ⇒ IP Address: Automatically Assigned
- ⇒ Router/Gateway: 192.168.1.1

❖ **Static IP Settings Applied:**

- IP Address: 192.168.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 8.8.8.8

❖ **Testing Result:**

After assigning the static IP address, the device was able to connect to the internet successfully. Websites opened normally, which confirmed that the static IP configuration was working.

❖ **After Testing:**

The IP settings were changed back to Obtain IP Address Automatically (DHCP) to prevent future network problems.

- 4) In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

- If two devices are assigned the same IP address, an IP address conflict occurs. One or both devices may lose network or internet access, causing connection problems until each device has a unique IP address.